





Why does
it matter?

Why should YOU do it?



**54% of Americans rely on
general news sources for
scientific information.**

Science & Engineering Indicators, B.G. Southwell & K. White (2022)



So we need to populate it with smart, accurate science.



34% of Americans think the
impact of science on society is
equally positive and negative

Public Views on Science and Society, Pew Research Center (2023)



8% said entirely negative; this percentage is up since 2021: 28% & 7% (respectively)
Adding to the quality and clarity of science on the internet might also help people to
think that scientist are more important

A crowded PhD market

Research doctorate recipients from U.S. colleges and universities: 1958–2022

(Number and percent)

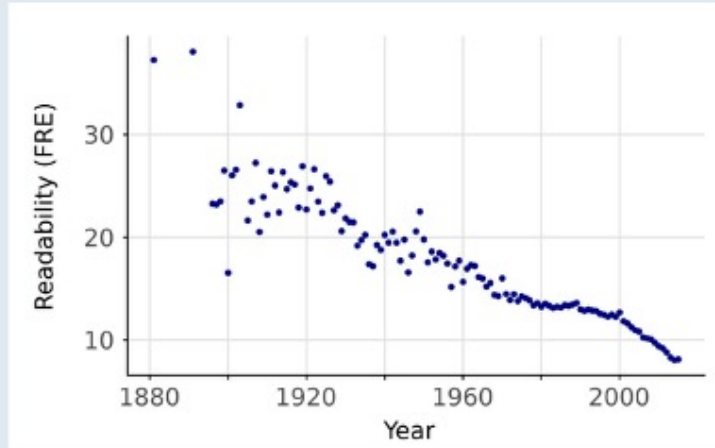
Year	Doctorate recipients	% change from previous year
2009	49,552	1.6
2010	48,028	-3.1
2011	48,909	1.8
2012	50,943	4.2
2013	52,703	3.5
2014	53,986	2.4
2015	54,886	1.7
2016	54,809	-0.1
2017	54,552	-0.5
2018	55,078	1.0
2019	55,609	1.0
2020	55,208	-0.7
2021	52,194	-5.5
2022	57,596	10.3

* = value between -0.05% and 0.05% and not equal to 0.0%.

National Center for Science & Engineering Statistics, Survey of Earned Doctorates (2022)

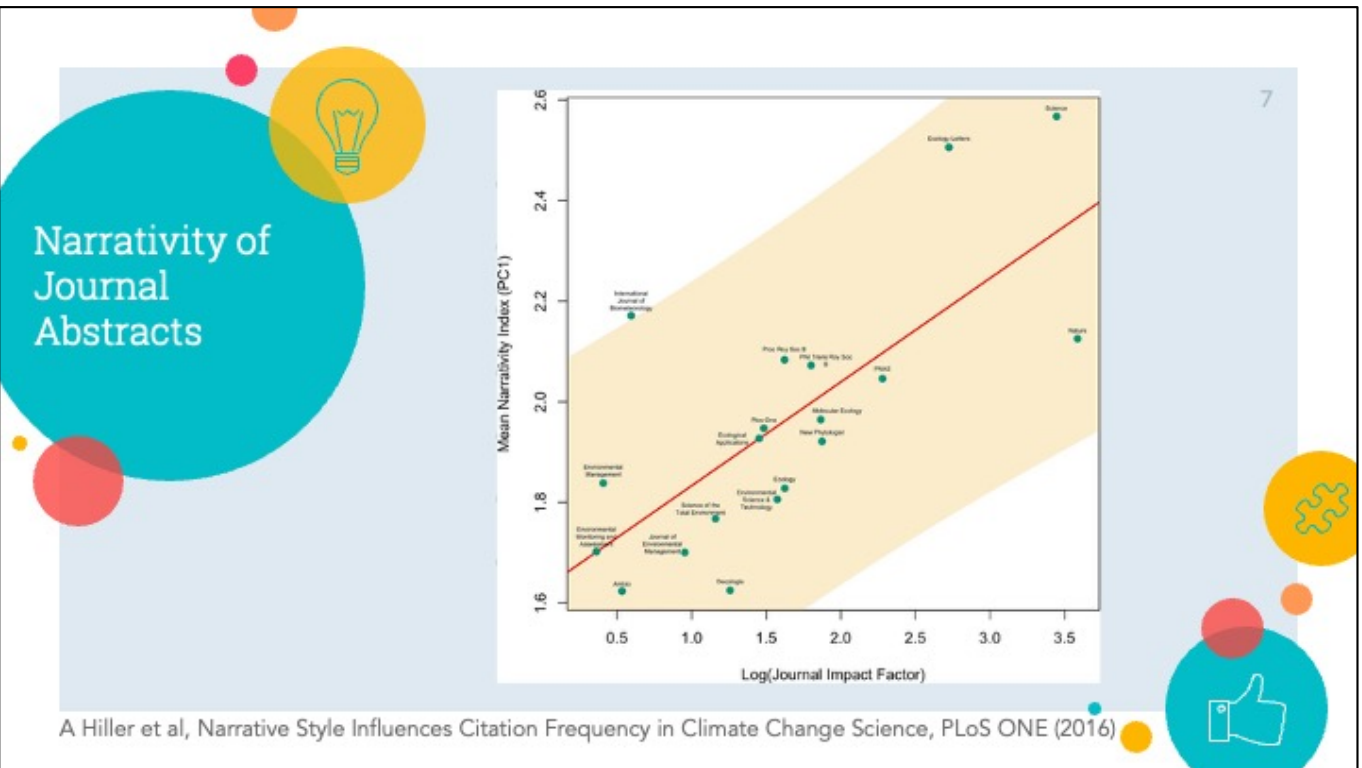
Selfish reasons. As Aaron mentioned yesterday, writing and communicating the results of your research matter a lot to the ability to get jobs

Readability of Scientific Journal Articles



Plaven-Sigray, Matheson, Schiffler, Thompson (2017)

And while we know that, overall, the “readability” of academic writing is declining



It is very much the case that the more narrative (code for understandable) your science is, the higher the likelihood that it will be cited by others, which (in academia) is an important metric for success



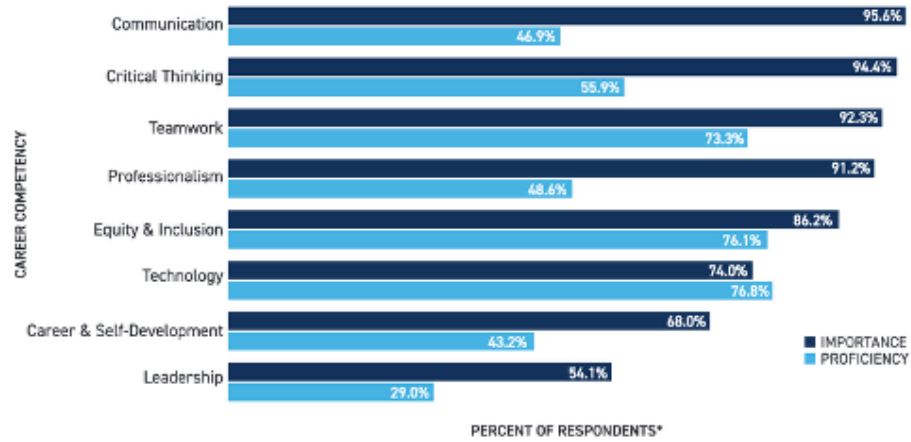
National Survey
of Employers
Priorities

#1: Communication with persons inside and outside the organization

National Association of Colleges and Employers Survey (2022)

And, even if you don't plan to remain in academia, being able to communicate is essential. Indeed, learning to communicate and clear and accessible ways about complex topics may be even more essential outside of the academy. (Interestingly, it is listed as #1 (4.49 on a 5 pt scale) out of 8 career competencies, but when rating recent grads on there competencies it comes in #5 (3.5 on that same scale)

FIGURE 46: IMPORTANCE VS. PROFICIENCY ON CAREER READINESS COMPETENCIES, BY PERCENT OF RESPONDENTS



*The percentages corresponding to "importance" represent, among all responding employers, the percentage that, on a five-point scale, indicated that the respective competency was either "very important" (4) or "extremely important" (5) for college graduates to have to enter their workforce. The percentages corresponding to "proficiency" represent, among all responding employers, the percentage that, on a five-point scale, rated recent graduates either "very" (4) or "extremely" (5) proficient in the respective competency.



So how do we
learn to do it?

In 1990, Gopen & Swan published an essential article on scientific writing.

American Scientist online
The Magazine of Science for the Modern World

The Science of Scientific Writing

If the reader is to grasp what the writer means, the writer must understand what the reader needs.
George A. Gopen, Judith A. Swan

This article was originally published in the November/December 1990 issue of American Scientist.

Science is often hard to read. Most people assume that its difficulties are born out of necessity, out of the extreme complexity of scientific concepts, data and methods. We argue here that complexity of thought need not lead to impenetrability of expression: we demonstrate a number of rhetorical principles that can produce clarity in communication without oversimplifying scientific issues. The results are adaptive, not merely cosmetic: improving the quality of writing actually improves the quality of thought.

The fundamental purpose of scientific discourse is not the mere presentation of information and thought, but rather its actual communication. It does not matter how pleased an author might be to have converted all the right data into sentences and paragraphs; it matters only whether a large majority of the reading audience accurately perceives what the author had in mind. Therefore, in order to understand how best to improve writing, we would do well to understand better how readers go about reading. Such an understanding has recently become available through work done in the fields of rhetoric, linguistics and cognitive psychology. It has helped to produce a methodology based on the concept of reader expectations.

Writing with the Reader in Mind: Expectation and Context

Readers do not simply read; they interpret. Any piece of prose, no matter how short, may "mean" in 10 (or more) different ways to 10 different readers. This methodology of reader expectations is founded on the recognition that readers make many of their most important interpretive decisions about the substance of prose based on clues they receive from its structure.

This interplay between substance and structure can be demonstrated by something as basic as a simple table. Let us see that in tracking the temperature of a liquid over a period of time, an investigator takes measurements every three minutes and records a list of temperatures. These data could be presented by a number of written structures; here are two possibilities:

(Time)=10', °(temperature)=100, 94°, 7=20%, 94°, 7=20%, 94°, 7=20%, 94°, 7=20%, 94°, 7=20%, 94°

Time (min) temperature(°C)

0	100
3	94
6	78
9	71
12	62
15	50

I'm going to talk through its major points by using examples from their article. I highly recommend you apply these points to your own writing.

The smallest of the URF's (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH_2 -terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene has been identified as the animal equivalent of the recently discovered yeast H^+ -ATPase subunit 8 gene. The functional significance of the other URF's has been, on the contrary, elusive. Recently, however, immunoprecipitation experiments with antibodies to purified, rotenone-sensitive NADH-ubiquinone oxido-reductase [hereafter referred to as respiratory chain NADH dehydrogenase or complex I] from bovine heart, as well as enzyme fractionation studies, have indicated that six human URF's (that is, URF1, URF2, URF3, URF4, URF4L, and URF5, hereafter referred to as ND1, ND2, ND3, ND4, ND4L, and ND5) encode subunits of complex I. This is a large complex that also contains many subunits synthesized in the cytoplasm.

What makes this passage difficult? Everyone will say jargon. Many will also say something about specialized knowledge.

(Note to readers: URF stands for "Uninterrupted Reading Frame," which describes a segment of DNA organized in a way that means it can encode a protein)

The smallest of the URF's, and [A], has been identified as a [B] subunit 8 gene. The functional significance of the other URF's has been, on the contrary, elusive. Recently, however, [C] experiments, as well as [D] studies, have indicated that six human URF's [1-6] encode subunits of Complex I. This is a large complex that also contains many subunits synthesized in the cytoplasm.

Same passage, but confusing jargon has (mostly) been eliminated. Questions, however, remain:

What does the first sentence have to do with the last sentence?

Does the third sentence contradict what the second sentence says (i.e., is the functional significance of URFs still elusive)?

Will the passage lead to further discussion about URFs, about Complex 1, both?

Even if we know the meaning of all of the technical terms, these things are still confusion.

Subject-Verb Separation

The **smallest** of the URF's (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH₂-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene **has been identified** as the animal equivalent of the recently discovered yeast H⁺-ATPase subunit 8 gene.

Part of the problem with easily comprehending the passage is related to subject-verb separation. In the original first sentence, the grammatical subject “the smallest” is separated from the verb “has been identified” by 23 words.

Subject-Verb Separation

The smallest of the URF's (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH₂-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene **has been identified** as the animal equivalent of the recently discovered yeast H⁺-ATPase subunit 8 gene.

The smallest of the URF's is URFA6L, a 207-nucleotide (nt) reading frame overlapping out of phase the NH₂-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene; **it has been identified** as the animal equivalent of the recently discovered yeast H⁺-ATPase subunit 8 gene.

A possible revision is on the right, here.

Stress-Position

Don't attempt to emphasize more than one important piece of information in a sentence. And highlight that information by placing it at the beginning or end of a sentence.

However, moving the subjects and verbs around may make the passage we read easier to process, but it still doesn't clear up all of the ambiguities I mentioned earlier. For that, we need to disambiguate some of the knotty sentences and take advantage of the stress position.

Stress-Position

Recently, however, immunoprecipitation experiments with antibodies to purified, rotenone-sensitive NADH-ubiquinone oxidoreductase [hereafter referred to as respiratory chain NADH dehydrogenase or complex I] from bovine heart, as well as enzyme fractionation studies, have indicated that six human URF's (that is, URF1, URF2, URF3, URF4, URF4L, and URF5, hereafter referred to as ND1, ND2, ND3, ND4, ND4L, and ND5) encode subunits of complex I.

Recently, however, several human URF's have been shown to encode subunits of rotenone-sensitive NADH-ubiquinone oxidoreductase. This is a large complex that also contains many subunits synthesized in the cytoplasm; it will be referred to hereafter as respiratory chain NADH dehydrogenase or complex I. Six subunits of Complex I were shown by enzyme fractionation studies and immunoprecipitation experiments to be encoded by six human URF's (URF1, URF2, URF3, URF4, URF4L, and URF5); these URF's will be referred to subsequently as ND1, ND2, ND3, ND4, ND4L and ND5.

A possible revision is on the right, here. You can see how this revision breaks up the information in what was the third sentence (left) of that passage and puts it in three sentences. This makes it easier to see where particular ideas begin and end and how they relate to each other.



Calcium blockers can control muscle spasms. Sarcomeres are the small units of muscle fibers in which these drugs work. Two filaments, one thick and one thin, are in each sarcomere. The proteins actin and myosin are contained in thin filament. When actin and myosin interact, your heart contracts.



Let's look at another passage to talk more about the stress position and its importance. Generally speaking, this passage is much easier to understand than the other in terms of the vocabulary and jargon it uses.

Calcium blockers can control muscle spasms. **Sarcomeres** are the small units of muscle fibers in which these drugs work. **Two filaments**, one thick and one thin, are in each sarcomere. **The proteins actin and myosin** are contained in thin filament. When **actin and myosin** interact, your heart contracts.

However, when we look at the topics of each sentence in isolation, what do we notice? Many of these concepts/words are appearing for the very first time in these positions. This is a problem for a reader who likes to be introduced to new ideas/concepts by relating them to old ones.

Stress-Position

Put old information at the beginning of sentences, new information at the end.

This leads us to a new rule.

Stress-Position

Calcium blockers can control muscle spasms. Sarcomeres are the small units of muscle fibers in which these drugs work. Two filaments, one thick and one thin, are in each sarcomere. The proteins actin and myosin are contained in thin filament. When actin and myosin interact, your heart contracts.

Muscle spasms can be controlled with drugs known as calcium blockers. Calcium blockers work in small units of muscle fibers called sarcomeres. Each sarcomere has two filaments, one thick and one thin. The thin filament contains two proteins, actin and myosin. When actin and myosin interact, your heart contracts.

In the revised version, we can see how using known information makes the entire passage easier to parse.



Transcription of the 5S RNA genes in the egg extract is TFIIIA-dependent. This is surprising, because the concentration of TFIIIA is the same as in the oocyte nuclear extract. The other transcription factors and RNA polymerase III are presumed to be in excess over available TFIIIA, because tRNA genes are transcribed in the egg extract. The addition of egg extract to the oocyte nuclear extract has two effects on transcription efficiency. First, there is a general inhibition of transcription that can be alleviated in part by supplementation with high concentrations of RNA polymerase III. Second, egg extract destabilizes transcription complexes formed with oocyte but not somatic 5S RNA genes.



Let's now look at one final passage to illustrate a couple more principles to keep in mind as you write. As with previous passages, this one is confusing for a whole host of reasons. The positions of topics in the sentence do not allow us to understand which of the many subjects the passages raises that we should focus on. Moreover, the verbs are weak, which also inhibits us from best understanding the actions.

Focus

Make a single topic the focus of a paragraph whenever possible. Use the stress positions to emphasize that establish that focus.

This leads us to a new rule.

Verbs

Use strong verbs that indicate the action that you aim to express.

This leads us to a new rule.

Focus & Verb Position

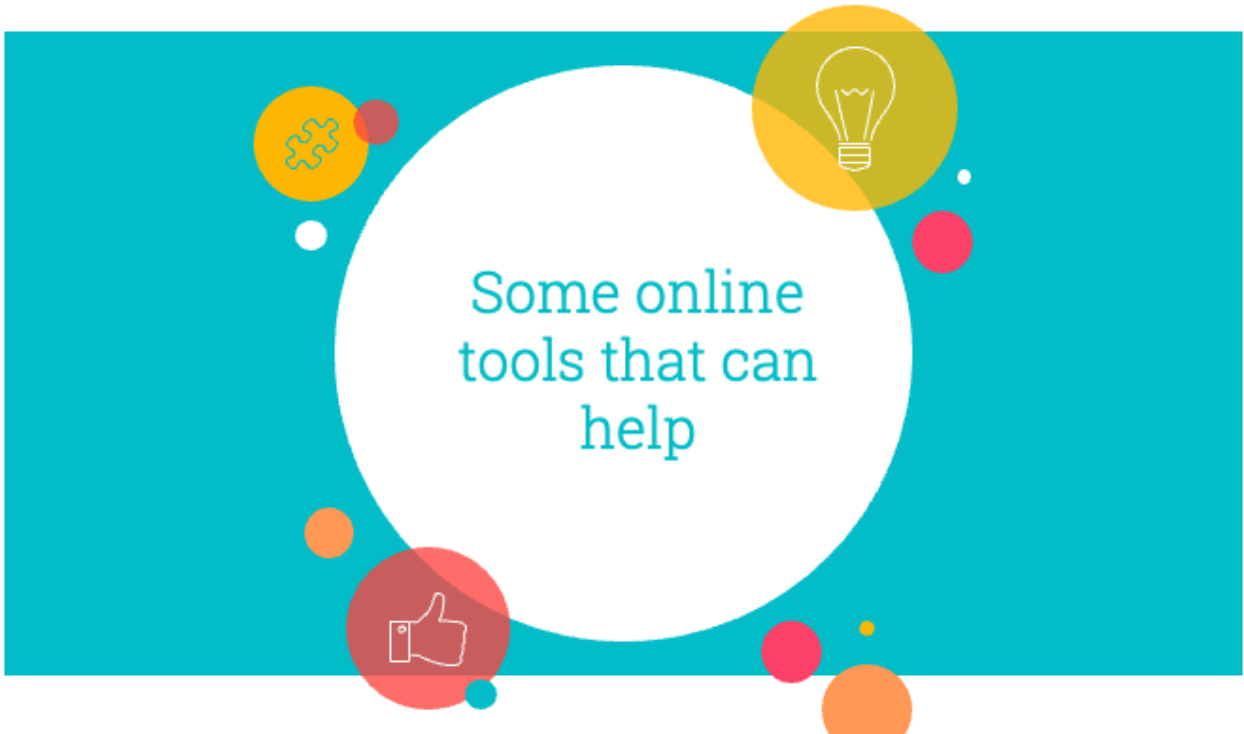
Transcription of the 5S RNA genes in the egg extract is TFI_{IIA}-dependent. This is surprising, because the concentration of TFI_{IIA} is the same as in the oocyte nuclear extract. The other transcription factors and RNA polymerase III are presumed to be in excess over available TFI_{IIA}, because tRNA genes are transcribed in the egg extract. The addition of egg extract to the oocyte nuclear extract has two effects on transcription efficiency. First, there is a general inhibition of transcription that can be alleviated in part by supplementation with high concentrations of RNA polymerase III. Second, egg extract destabilizes transcription complexes formed with oocyte but not somatic 5S RNA genes.

In the egg extract, the availability of TFI_{IIA} limits transcription of the 5S RNA genes. This is surprising because the same concentration of TFI_{IIA} does not limit transcription in the oocyte nuclear extract. In the egg extract, transcription is not limited by RNA polymerase or other factors because transcription of tRNA genes indicates that these factors are in excess over available TFI_{IIA}. When added to the nuclear extract, the egg extract affected the efficiency of transcription in two ways. First, it inhibited transcription generally; this inhibition could be alleviated in part by supplementing the mixture with high concentrations of RNA polymerase III. Second, the egg extract destabilized transcription complexes formed by oocyte but not by somatic 5S genes.

In the revised version, we can see how getting TFI_{IIA} into the topic position whenever possible brings a bit more focus. Even more, we can see how making the verbs stronger – emphasizing limiting and inhibiting – provide focus to the passage that helps the reader to understand its importance.

Summary of principles

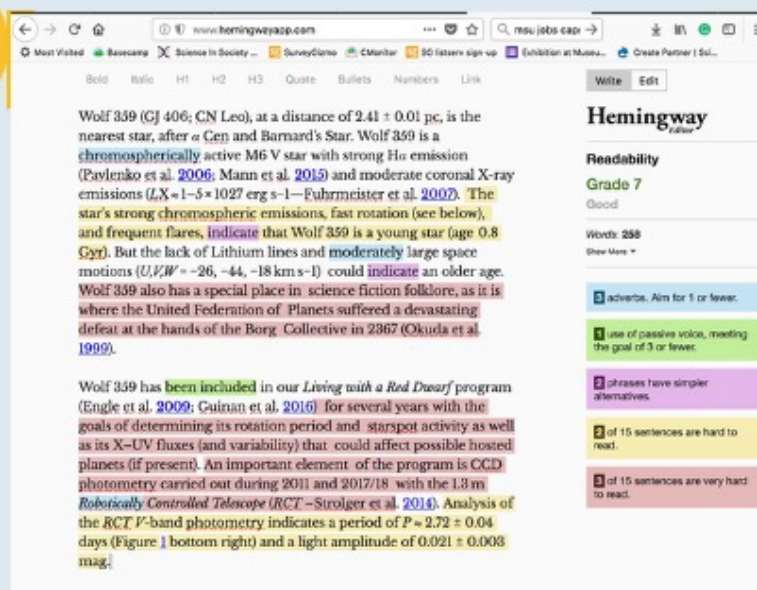
- Place subjects & verbs as closely together as possible
- Use stress positions to introduce new topics/concepts, establish relationships between topics, and establish a primary topic for a given paragraph
- Use strong verbs to emphasize your points



Some online
tools that can
help

Hemingway App

www.hemingwayapp.com



Wolf 359 (GJ 406; CN Leo), at a distance of 2.41 ± 0.01 pc, is the nearest star, after α Cen and Barnard's Star. Wolf 359 is a **chromospherically** active M6 V star with strong H α emission (Pavlenko et al. 2006; Mann et al. 2015) and moderate coronal X-ray emissions ($L_X = 1\text{--}5 \times 10^{27}$ erg s $^{-1}$ —Fuhrmeister et al. 2007). The star's strong chromospheric emissions, fast rotation (see below), and frequent flares, **indicate** that Wolf 359 is a young star (age 0.8 Gyr). But the lack of Lithium lines and **moderately** large space motions ($U, V, W = -26, -44, -18$ km s $^{-1}$) could **indicate** an older age. Wolf 359 also has a special place in science fiction folklore, as it is where the United Federation of Planets suffered a devastating defeat at the hands of the Borg Collective in 2367 (Okuda et al. 1999).

Wolf 359 has **been included** in our *Living with a Red Dwarf* program (Engle et al. 2009; Guinan et al. 2016) for several years with the goals of determining its rotation period and starspot activity as well as its X-UV fluxes (and variability) that could affect possible hosted planets (if present). An important element of the program is CGD photometry carried out during 2011 and 2017/18 with the 1.3 m *Robotically Controlled Telescope (RCT)*—Stroilger et al. 2014). Analysis of the *RCT* *V*-band photometry indicates a period of $P = 2.72 \pm 0.04$ days (Figure 1 bottom right) and a light amplitude of 0.021 ± 0.003 mag.

Hemingway
Editor

Readability
Grade 7
Good

Words: 258
Show More

- 1 adverbs. Aim for 1 or fewer.
- 1 use of passive voice, meeting the goal of 3 or fewer.
- 2 phrases have simpler alternatives.
- 2 of 15 sentences are hard to read.
- 2 of 15 sentences are very hard to read.

Hemingway was famous for having terse, clear, active prose. This app uses some of these principles to help show you sentences that might be difficult to read and provides you with alternatives

De-Jargonizer

scienceandpublic.com

The screenshot shows a web browser window with the URL 'scienceandpublic.com'. The page has a dark header with a 'Start' button. The main content area is titled 'Result' and displays a paragraph of text about 'Wolf 359'. The text is color-coded: 'Wolf 359' is orange, 'CN Leo' is red, 'nearest star' is green, 'alpha Centauri' is blue, 'chromospherically active' is red, 'M8 V star' is green, 'strong H-alpha emission' is red, 'Parker et al. 2006' is blue, 'Mann et al. 2010' is green, 'moderate coronal X-ray emissions' is red, 'LX=1-6x10^27 erg s^-1' is blue, 'Patzelt et al. 2007' is green, 'The star's strong chromospheric emissions, fast rotation (see below), and frequent flares, indicate that Wolf 359 is a young star (agelessim 0.8 Gyr)' is red, 'But the lack of Lithium lines and moderately large space motions (U, V, W=-26, -44, -19 km s^-1)' is green, 'could indicate an older age' is red, 'Wolf 359 also has a special place in science fiction folklore, as it is where the United Federation of Planets suffered a devastating defeat at the hands of the Borg Collective in 2387 (Dixie et al. 1999)' is blue.

Below the text, there is a section titled 'Wolf 359 has been included in our Living with a Red Dwarf program (Engle et al. 2008; Guinan et al. 2016) for several years with the goals of determining its rotation period and starspot activity as well as its X-ray fluxes (and variability) that could affect possible hosted planets (if present). An important element of the program is CCD photometry carried out during 2011 and 2017/18 with the 1.3m Robotically Controlled Telescope (RCT - Stroger et al. 2014). Analysis of the RCT V-band photometry indicates a period of P = 2.72 ± 0.04 days (Figure 1 bottom right) and a light amplitude of 0.021 ± 0.003 mag.'

On the right side of the result, there is a rainbow-colored arc graphic and a table of statistics:

Common:	72%, 159
Mid-Frequency:	16%, 35
Rare:	12%, 26
Suitability for general audience score:	80
Number Of Words:	220

Below the table is a 'Download' button and a 'Share' link.

This helps to identify jargon, though doesn't really help you see solutions. You can also see from this example that names and other things that really aren't jargon get identified as jargon by these bots. So, you always need to be a human when you parse them.



Helpful reading

- The Science of Scientific Writing, Gopen & Swan
- Writing Science in Plain English, Anne Greene
- Scientific Writing & Communication, Angelika Hofmann





Thanks!

Any questions?

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